

Stiffness in the Robertson Chemical Kinetics System

What explicit methods pay — what implicit methods buy

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Numerical Analysis of ODEs and PDEs — Project 1

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One question: *how does stiffness translate into a practical step-size restriction for explicit methods — and what does an implicit method pay for the same accuracy?*

The problem: Robertson chemical kinetics

$$\begin{aligned}y_1' &= -0.04 y_1 + 10^4 y_2 y_3 \\y_2' &= 0.04 y_1 - 10^4 y_2 y_3 - 3 \times 10^7 y_2^2 \\y_3' &= 3 \times 10^7 y_2^2\end{aligned} \quad y(0) = [1, 0, 0]^T, \quad 0 \leq t \leq 40$$

- ▶ Rate constants span 10^8 ; the state spans 10^5 ($y_2 \sim 10^{-5}$ vs $y_1 \sim 1$)
- ▶ Invariant plane $y_1 + y_2 + y_3 = 1 \Rightarrow$ structural zero eigenvalue
- ▶ Stiffness ratio over the two nonzero eigenvalues grows to 1.584×10^5 at $t = 40$

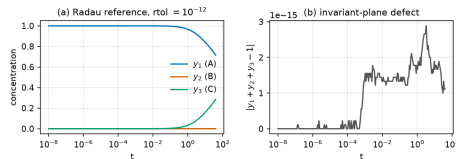
Takeaway: a benchmark designed so that explicit methods fail for a measurable, quantitative reason.

What we built — and how we trust it

From scratch: explicit Euler, RK4, implicit Euler + Newton, adaptive step-doubling RK4 — swappable f , built-in cost counters.

Our own oracle: scipy Radau, $\text{rtol} = 10^{-12}$, tightened $\sim 45\times$; digits that survive the repeat are the only ones we quote.

Takeaway: $y(40) = (0.7158271, 9.1855 \times 10^{-6}, 0.2841637)$ to 15 digits; invariant defect 2.9×10^{-15} ; Problem Pack check value reproduced.



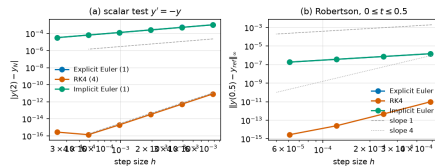
Orders are verified, not assumed

Observed order (slope of $\log err$ vs $\log h$) on the stable window $0 \leq t \leq 0.5$:

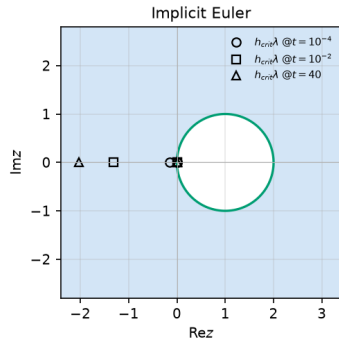
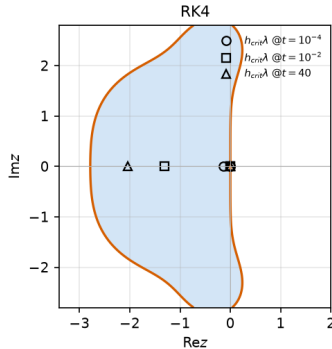
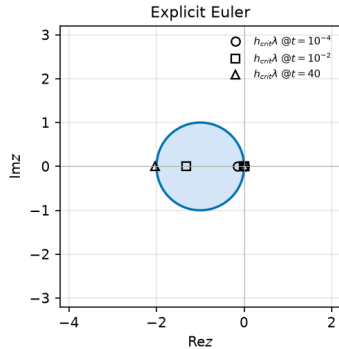
	Euler	RK4	Impl.
scalar	1.00	4*	1.00
Robertson	1.00	4.35	1.00

* round-off floor flattens the tail.

Takeaway: Euler and implicit Euler have the same order-1 constant — they differ in stability, not per-step accuracy.



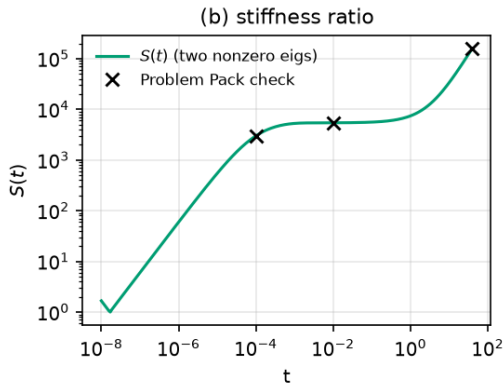
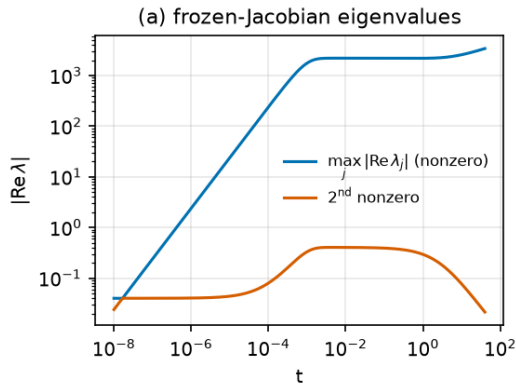
Stability regions: hard ceilings for explicit methods



$h_{\text{Euler}} < 2/|\lambda|_{\text{max}} = 0.00059$, $h_{\text{RK4}} < 2.8/|\lambda|_{\text{max}} = 0.00083$, implicit Euler A-stable.

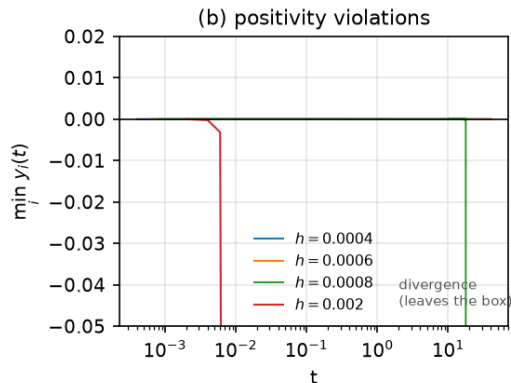
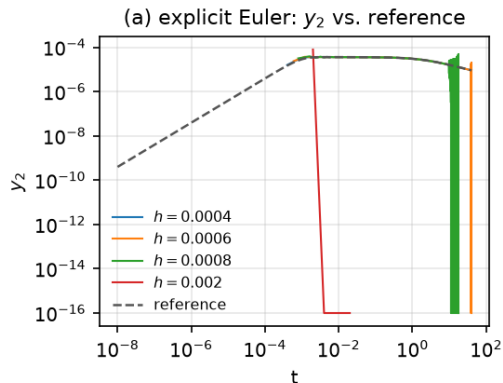
Takeaway: the frozen-Jacobian eigenvalue at $t = 40$ sets a hard step ceiling — and we test it empirically next.

Stiffness grows along the trajectory



Takeaway: $S(t)$: $3.03 \times 10^3 \rightarrow 5.42 \times 10^3 \rightarrow 1.584 \times 10^5$ at $t = 10^{-4}, 10^{-2}, 40$ — matches the Problem Pack check values.

Explicit failure: physics fails before arithmetic



measured threshold $h_{\text{crit}} = 0.0006$ vs. prediction 0.00059 (2% error).

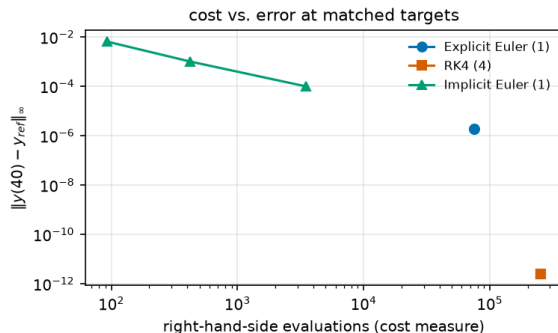
Takeaway: positivity is lost (bounded but non-physical) *before* blow-up — “did not explode” is not “is right”.

Cost vs. accuracy at matched error

Method	Target	h	error	nfev	wall
Explicit Euler	10^2	0.00053	1.9×10^{-6}	75,395	0.50 s (stab.-pinned)
Explicit Euler	10^3	0.00053	1.9×10^{-6}	75,395	0.51 s (stab.-pinned)
Explicit Euler	10^4	0.00053	1.9×10^{-6}	75,395	0.47 s (stab.-pinned)
RK4	10^2	0.00063	2.6×10^{-12}	252,060	2.11 s (stab.-pinned)
RK4	10^3	0.00063	2.6×10^{-12}	252,060	2.50 s (stab.-pinned)
RK4	10^4	0.00063	2.6×10^{-12}	252,060	2.07 s (stab.-pinned)
Implicit Euler	10^2	2	0.0065	93	0.01 s
Implicit Euler	10^3	0.3	0.001	421	0.03 s
Implicit Euler	10^4	0.029	0.0001	3,505	0.22 s

Takeaway: implicit Euler wins by $810 \times / 179 \times / 21 \times$ at targets $10^{-2}/10^{-3}/10^{-4}$ — explicit methods are pinned at their stability limit and cannot loosen up.

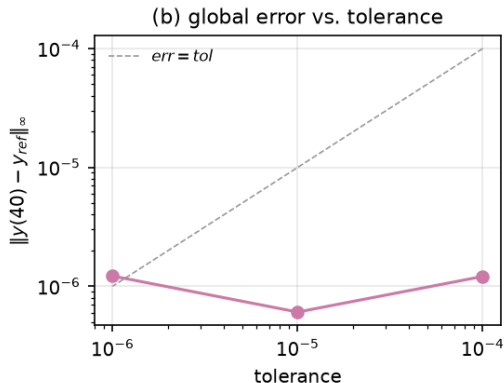
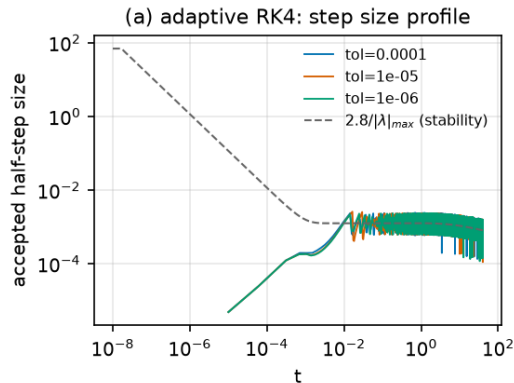
RK4's order is wasted on the stiff interval



- ▶ RK4: error 2.6×10^{-12} *always* — its h is pinned at the stability limit for every target
- ▶ highest nfev, accuracy far beyond anything requested

Takeaway: on a stiff interval the binding constraint is stability, not accuracy — order alone does not win.

Adaptive RK4: accuracy responds — stability does not



step size adapts over 3 orders of magnitude; controller tracks the $2.8/|\lambda|_{\max}$ envelope from below.

Takeaway: error follows the tolerance down to $\sim 10^{-6}$ and then *plateaus*: adaptivity buys accuracy, not stability.

Conclusions

1. frozen-Jacobian prediction verified to $\sim 2\%$ (0.0006 vs 0.00059)
2. at matched error, implicit Euler wins by $21\times-810\times$ (nfev) on $[0, 40]$
3. RK4's order 4 is irrelevant here — stability pins its step
4. adaptive controller: error plateaus at the stability ceiling
5. diagnostics fail in order: positivity $\rightarrow$ stability $\rightarrow$ (conservation survives)

Takeaway: stiffness is a stability story — measure it, don't assume it.

Reproducibility

- ▶ `python code/run_all.py` regenerates every figure and every number (`results/results.json` → `numbers.tex` → report)
- ▶ `requirements.txt`: `numpy`, `scipy`, `matplotlib` — no other dependencies
- ▶ AI use documented in the AI Transparency Log (report App. B); ICS backed by git history (App. A)

Repository: github.com/2360751/robertson-stiff-ode — all team members can explain and modify every part (viva-ready).